19. Greater the 'a' value, easier the liquefaction

21. $an^2/V^2 = atm$

$a = atm\ L^2/mol^2 = L^2\ mol^{-2}\ atm$

$nb = L$

$b = L/mol = L\ mol^{-1}$

Option (c)

22. **Correct Statement:** Critical temperature of CO_2 is 304 K. It means that CO_2 cannot be liquefied above 304 K, whatever the pressure may applied.

Pressure is inversely proportional to volume

Option (d)

23.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$= \frac{m}{\left(\frac{nRT}{P}\right)} = \left(\frac{m}{n}\right)\frac{P}{RT}$$

$$= \text{Molar mass} \times \frac{P}{RT}$$

$$= \frac{28\ g\ mol^{-1} \times 5\ atm}{0.082\ L\ atm\ K^{-1}\ mol^{-1} \times 500\ K}$$

$$= 3.41 g\ L^{-1}$$

Option (c)

24. For a fixed mass of an ideal gas $V \alpha T$

$P \alpha 1/V$

and $PV = \text{Constant}$

Option (c)

25. At a given temperature and pressure

Volume α no. of moles

Volume α Mass / Molar mass

Volume α 28 / Molar mass

i.e. if molar mass is more, volume is less. Hence HI has the least volume

Option (d)

II Key to short answer questions:

44. $T_1 = 15^0\ C + 273$ $T_2 = 38 + 273$

$T_1 = 288\ K$ $T_2 = 311\ K$

$V_1 = 2.58\ dm^3$ $V_2 = ?$

(P = 1 atom constant)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$V_2 = \left(\frac{V_1}{T_1}\right) \times T_2$$

$$= \frac{2.58\ dm^3}{288\ \cancel{K}} \times 311\ \cancel{K}$$

$V_2 = 2.78\ dm^3$ i.e. volume increased from $2.58\ dm^3$ to $2.78\ dm^3$

45. $V_1 = 8.5\ dm^3$ $V_2 = 6.37\ dm^3$

$T_1 = ?$ $T_2 = 0^0\ C = 273\ K$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$V_1 \times \left(\frac{T_2}{V_2}\right) = T_1$$

$$T_1 = 8.5\ \cancel{dm^3} \times \frac{273\ K}{6.37\ \cancel{dm^3}}$$

$T_1 = 364.28 K$

46. $n_A = 1.5\ mol$ $n_B = ?$

$V_A = 37.6\ dm^3$ $V_B = 16.5\ dm^3$

(T = 298 K constant)

$$\frac{V_A}{n_A} = \frac{V_B}{n_B}$$

$$n_A = \left(\frac{n_A}{V_A}\right) V_B$$

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